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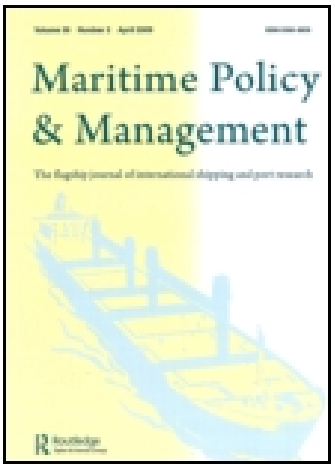

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Maritime sociology: towards a delimation of themes and analytical frameworks

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Maritime sociology: towards a delimitation of themes and analytical frameworks

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This paper sets out to evaluate the main areas of focus within maritime sociology and to link more closely than hitherto the various substantive discussions with general theoretical and conceptual approaches. More specifically, it suggests directions in which maritime sociology may be advanced at both empirical and conceptual levels in the future.

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1. Introduction

For upwards of a century there has been a recurrent, if at times fitful and sporadic, interest in the systematic study of a broad range of social phenomena within the maritime environment. Nevertheless, it remains the case that, in the substantial corpus of literature which now obtains in this respect, the analytical premises have generally been circumscribed and, above all, have been insufficiently integrated into central themes within sociology itself. By the same token, too, in actual research studies there has seldom been any rigorous attempt, within advanced operational designs, to formulate precise models to test the significance of major propositions in general currency.

It is thus the objective of this contribution to conduct an inquiry into the current state of development of maritime sociology and, more specifically, to suggest directions in which in the forthcoming years this area of applied scholarship may be fruitfully advanced at both empirical and conceptual levels. Furthermore, although in the outline which follows it is not intended to present merely a bibliographic review of salient studies, the considerable number of references which will be accommodated are planned to be of pedagogic as well as research interest. The examination will thus proceed by evaluating various problems of definition and of delimitation of this branch of inquiry and, then, prior to a discourse on future strategy, the main focal themes for analysis will be adumbrated under the following headings:

(*a*) The concept of a social system.
(*b*) Community and association.
(*c*) Social action and culture.
(*d*) Structure, differentiation and stratification.
(*e*) Processes (including the patterns of recruitment and socialization).
(*f*) Change with special reference to the impact of technology.
(*g*) Conflict and deviance.

2. Definition and delimitation of the area of study

Maritime sociology has been typically defined in terms of 'identifying in specific cases, the social and cultural forces which determine the specific definitions of the marine environment which shape its use and exploitation' [1]. On the face of it, such an interpretation is unexceptionable and, indeed, the focus upon both the use and exploitation of maritime resources is an especially valuable point of departure.

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Nevertheless, on more detailed inspection two main problems stubbornly persist: the nature and character of relevant social and cultural forces; and the precise social groups and relationships to be incorporated in this area of study itself.

To begin with, therefore, although ultimately the saliency or otherwise of given propositions can only be settled by empirical researches, the degree of integration of concepts in maritime sociology with classical definitions and interpretations in the parent discipline is a matter for some concern. Indeed, the bulk of familiar terms such as total institution, community, deviant behaviour and so on have, by and large, been developed in an *ad hoc* and inductive manner. But while up to a point this is a reasonable *modus operandi* in the early stages of development of an area of inquiry, if maritime sociology is ever to become sophisticated and conceptually adequate, such analyses must be supplemented, and even ultimately wholly replaced, by more deductive modes of inquiry in which mainstream concepts are explicitly applied and tested. Again, although social and cultural phenomena are clearly within the domain of the sociologist, the implications of economic changes, governmental policies, legislative enactments and technological movements for relationships between maritime personnel are also legitimate themes for investigation.

By the same token, too, the precise groups encompassed in this area of study are by no means wholly unambiguous. After all, while the officers and crew (and the technical specialists) on board ship engaged in transportation (seamen) and extraction (fishermen) are obviously relevant, the inclusion of dockworkers (longshoremen) and shipbuilders, while on balance defensible, is clearly open to considerable dispute. Moreover, other groups which await detailed investigation include those engaged in oil and gas extraction from the seabed, lighthouse operators and the crew of lifeboats. Similarly, the relationships between various maritime groups and shore-based personnel (families, shipping companies, trade unions and so on) should clearly be accommodated.

3. The concept of a social system

Turning, then, to actual sociological analyses of maritime phenomena, in the formative literature it was a familiar point of departure to conceive of social relationships on board ships in terms of social systems [2]. In essence, such approaches began by the delineation of various roles and functions which were seen as mutually interdependent and reinforcing and as being co-ordinated through administrative and control structures based on hierarchical modes of authority. Moreover, the interlocking social structures on board ship were interpreted in terms of serving certain manifest purposes (transportation, extraction and so on). In a comprehensive vein, therefore, Aubert, for instance, viewed the ultimate objective of ships as production and the social patterns as encompassing such further dimensions as proximity, complementarity, similarity, recruitment, socialization, stratification, normative orientations and conflict and deviance. Again, in cognate formulations, the crew itself was analysed by reference to system concepts [3].

Nevertheless, by virtue of constituting essentially 'grounded theories' or frameworks generated in empirical discourses, these perspectives of life on board ship were seldom couched in more general types of structural functionalist analysis. Certainly there was little evidence of the emphasis in classical sociological theory on the problem of order which, in this school of thought, had to be solved in all social systems by means of the specialized activities of adaptation (to the wider environment), goal attainment (the accomplishment of given objectives), integration (covering the relationship between social units) and pattern maintenance and tension management. Again the

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structure of social systems in classical sociology encompassed: (*a*) subgroups of various types, interconnected by relational norms; (*b*) roles of disparate societies; (*c*) regulative norms governing subgroups and roles; and (*d*) cultural values [4]. Yet, while such concepts were discernible in early models of maritime life based on social system analogies, they were seldom developed in a systematic form and in a way which was directly linked with the dominant concerns of the parent discipline.

Naturally, in modern systems analysis of maritime phenomena, a number of advanced formulations have eventuated. Moreby has emphasized that the interdependence, or relationship, between parts constitutes an elemental definition of a system and has further stressed the importance of open systems including transactions with an environment. Moreover, not only did he envisage labour recruitment in shipping to be characterized by a 'disturbed, reactive environment', but also observed that the behaviour of individual seafarers and groups could be analysed in terms of socio-technical systems. Similarly, in his view, systems thinking in shipping could be understood in respect of three interlocking patterns comprised from the disparate economic, technical and social elements which are germane to this area of study [5]. To be sure, the concept of a 'total institution' has proved to be particularly fruitful in the maritime context and this was undoubtedly developed directly from the sociological theories of Goffman [6]. Hence Aubert and Arner viewed ships in such terms and noted that relationships on board involved isolation (from shore-based communities), instability of the group on board, role differentiation and interaction barriers and an unbroken line of promotion [7]. By the same token, too, as Hopwood observed [8]:

> What is clear, however, is that a ship's crew constitutes a largely random selection of individuals, the majority of whom are unknown to each other until the moment of commencing a voyage. When the ship leaves port, these individuals are encapsulated by their ship-board environment, often for long periods of time; the internal social system of the ship is cut off from the wider society both by distance and by physical barriers. At sea, all aspects of life for the seamen are conducted in the same place with close scheduling of all phases of activity and, to a lesser extent, the same may be said of periods spent in port. The seaman lives at his place of work, and, in addition to his work periods, is forced to take his leave with fellow-workers and his superiors.

Again, Moreby has listed no less than 34 items covered by the total institution concept; a total which not only comprised the original Goffman components but also 17 different items deriving from maritime contexts themselves. The latter thus encompassed: lack of choice; territoriality; superficiality of personal contacts; lack of personal knick-knacks; identification and labelling by work roles; emotional under-nourishment; rejection of the sick as causing work overload on others; believed infallibility of decisions; greater tolerance of views; behavioural conformity; tendency to 'infantilize' subordinates by insecure in positions of authority; temporary ganging against outsiders; fantasy; avoidance of personal responsibility; irreversible nature of decisions; uncertainty without calling for certainty within (except in port where explosive disintegration of discipline may occur); lower members seen as irresponsible and incompetent (and, in some respects, systematically abused) [9].

Nevertheless, while the concept of a total institution has been commonly deployed for the analysis of maritime phenomena, its utility in this context has been open to a number of doubts and objections [10]. The voyage on board ship is relatively brief by comparison with, say, the situation which obtains among the inmates of mental asylums. Again, while discipline is more severe than in a manufacturing enterprise it is temporary and, in any event, has been subject to a number of changes over time. In the fishing industry there is also the prospect of rising from the ranks, which is not the

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situation for the prisoner. Moreover, with the advent of ship-board representation in merchant ships, sailors have increasingly gained opportunities for the redress of certain grievances during the actual voyage itself. Finally it should be emphasized that the concept of a 'total institution' is essentially a diagnostic term which is at best partially valid and also, like the social system frameworks, provides no real basis for explanation of key maritime phenomena or for changes in patterns of relationships both on board ship and with dependent shore-based occupations and social groups.

4. Community and association

Early literature on maritime sociology rested heavily on the distinction between community and association. To be sure, in many respects the implications of the notion of community in early studies in this area ran directly counter to the total institution idea since, rather than the personnel being conceived of as the equivalent of inmates who were previously unknown to one another, strong bonds of kinship and common affiliation based on similar modes of socialization were typically emphasized as the distinguishing characteristics of such diverse groups as fishermen, dockworkers and shipbuilders [11]. But the familiar sociological distinction between *gemeinschaft* (community) and *gesselschaft* (association) was to be interlinked in maritime studies, since the former was seen as a basis for the latter and as occasioning particularly the rise of strongly-bonded trade unions with traditions of militancy. Nevertheless, such observations differed from the arguments of classical sociological writers such as MacIver who observed that:

> A community is a focus of social life, the common living of social beings; an association is an organization of social life, definitely established for the pursuit of one or more common interests. An association is partial, a community is integral. Within a community there may exist not only numerous associations but also antagonistic associations [12].

The emergence of strongly bonded communities in maritime environments has been familiarily located in terms of the 'extreme' characteristics of the occupations which, as Tunstall has noted in the case of fishermen, involves the impact of work on non-work; exchange relations in fluctuating peer groups; an 'oppositional' pattern of leisure; a career structure based on early recruitment and early retirement; executive incomes, but proletarian identifications of skippers; the relationship between family, poverty and occupation; the 'misreading' of cues in recruitment, trade unionism and social values [13]. More generally, too, as Fricke has argued,

> a seafarer rarely lives and works alone... the seafarer, whether fisherman or merchant seaman, is at sea for economic reasons. His livelihood depends on the success of his occupation. He works and lives with others because this reduces the level of risk inherent in working alone, and adds to the capital available for exploitation of new technologies or markets, or the encouragement of existing status. The relationships derived from working and living with others, from engaging in interdependent activities, are social relationships, and form the basis for the development of an occupational community [14].

Again, as Hill has noted in the case of dockworkers, the traditional nature of the community has been maintained partly by 'the continued existence of elements of *gemeinschaft* society, so that primordial ties of kin, personal bonds between friends and a sense of moral obligation dominate the culture of the industry and its typical forms of organization' [15].

Moreover, in these diverse formulations, the saliency of community as a base for effective association has been typically propounded. This has been manifested above all

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in the familiar Kerr-Siegel hypothesis, in which the inter-industry propensity to strike was seen not only to be high amongst dockworkers (longshoremen) but also appeared to depend on the existence of a single occupational community, isolation of the group from workers in other industrial milieux, and homogeneity in terms of the lack of marked differentiation of tasks [16]. Similarly, Lockwood once argued that such communities tended to produce so-called 'proletarian traditionalism' in which: (*a*) the conception of social classes was viewed in dichotomous terms and as based on marked disparities of power; (*b*) career advancement was seen as either pointless or reprehensible; and (*c*) the typical orientation to work was solidaristic. In this last respect, then, relationships with superiors were defined as involving conflicts of interest; solidarity with workmates was valued and union organization was seen as a strength; work, while having an economic meaning, was viewed as a group activity and hence economic returns were at times sacrificed for solidarity and conformity with group norms; and ego involvement in work remained strong and was important for the satisfaction of workers' expressive and affective as well as instrumental 'needs' [17]. Moreover, while both the Kerr-Siegel hypothesis and the notion of proletarian traditionalism have been the subject of detailed criticism in subsequent empirical researches [18], this in no way invalidates the case that they were frequently deployed in early studies of maritime occupational groups or that they formed a strategic link between the concepts of community and association respectively.

5. Social action and culture

Even in the early studies of maritime occupations, however, it was fully appreciated that no comprehensive analysis could afford to ignore the subjective forces which shape social action and behaviour and, above all, the orientations and perceptions of the personnel concerned. Moreover, in a variety of interesting studies, the case was advanced that patterns of organization and decision-making, modes of training and so on could vary in different countries and in some measure reflect deeper cultural traits.

Taking first of all, then, the social-action perspectives, in a wide range of maritime literature the orientations and attitudes of the personnel concerned have of course been recorded and evaluated [19]. At times, too, the notion of work alienation including isolation, powerlessness, meaninglessness, and self-estrangement has surfaced [20]. Nevertheless, the degree of integration of these studies with the premises of this approach in the parent discipline has again proved to be far from satisfactory. Hence, analysts deploying the social-action perspective conceive of social systems as:

> the derivative of social action and interaction, a social world produced by its members, who are thus pictured as active, purposeful, self and socially-creative beings. The language of social action is thus the language of subjective meaning, in terms of which social actors define their lives, purposes, and situations; of the goals and projects they generate on the basis of their subjective meanings; of the means whereby they attempt to achieve their goals and realise their projects; of the social action upon which they embark in the prosecution of such attempts; of the social relationships into which they enter on the basis of their pursuit of goals and projects; and of social roles, institutions, and the social system conceptualized as the emergent product of their consequent social interaction [21].

More specifically, relating the Weberian types of social action [22] to the maritime situation would require a focus on the following:

(*a*) *Value-rational action:* the general ideologies and ethical values which inform social action. These include both the political and religious values of key maritime groups.

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(*b*) *Purposive or instrumental-rational action:* the orientations to personal projects involving power, status, material and career advancement and so on. The extent to which maritime personnel are influenced in their actions by such objectives.

(*c*) *Affectual conduct:* in a maritime context, this would include sentiments of belonging and loyalty to groups and to work-mates together with emotional attachments to maritime life itself.

(*d*) *Traditional action:* this is governed by past precedents and practices and most noticeably entails the socially transmitted cultural traits of different societies and social formations. In a maritime situation, these encompass the customs and practices of diverse communities, associations and groups.

Again, insofar as culture is concerned, its impact has been discussed in a variety of different studies. To begin with, it would appear that training patterns in some measure reflect culture, since while in the United Kingdom the shipping companies would appear to be the primary beneficiaries of such practices, in the United States it is the individuals concerned and in Spain the central government [23]. By the same token, in Italy not only have communities of fishermen been governed by regulations which can be traced back to the 15th century; but also, the development of co-operative modes of organization has been a traditional departure influenced by religious values [24]. Again, even in the field of technological innovation, the typically slower pattern of adaptation in the British context has not only been identified in studies of social history, but may also be traced to general patterns and traits embodied in the wider culture [25]. Similarly, at a more specific level, Sharpe has argued for a change in the central employment philosophy in UK merchant navy manpower systems, even though he has also noted the far-reaching consequences of a wider cultural system which has encouraged an occupational structure based on so-called 'role-fillers' (i.e. flexible roles with limited skills which can be easily occupied). This situation has been encouraged by the absence of stability in the composition of the ship's company, a 'need' for centrally established measures of role competence permitting safe interchangeability of persons to fill roles adequately on any ship, a sufficient flow of acceptable entrants for each role, the volatile nature of ship and cargo markets, and most ships being dependent upon the same on-board role structure for effective and safe operation [26].

6. Structure, differentiation and stratification

The concept of social structure has also proved to be of considerable significance for the analysis of maritime phenomena. To some extent, as we have earlier observed, this notion has been integrated into the various frameworks and approaches based on social system analogies; but it has also been related to patterns of differentiation and stratification on board ship. Indeed, in all societies and large-scale organizations some degree of stratification may be observed, a situation which implies divergencies in status and social esteem, in material and economic circumstances and in decision-making authority and power of the personnel in question. Nevertheless, the principles of stratification clearly differ, while the patterns which obtain even in quite similar occupations are by no means wholly uniform. Again, as may be illustrated from maritime environments, not only may dissimilar characteristics in the modes of stratification be observed at any given point in time, but also important variations may be detected historically as a result of bureaucratization, technical change and other salient forces influencing the nature of the respective social formations and structure concerned.

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In social system analysis, then, structure is typically related to function and in merchant ships, for example as Hopwood has observed, the division of labour may thus be seen to encourage the specialization of activities into four main sections or departments (deck, engine-room, catering and radio) [27]. But in stratificational approaches, emphasis is also placed on the observation that 'within each department, personnel are formally organized in a rigid hierarchy' [28]. Moreover, from a sociological point of view, two further issues are typically addressed: (*a*) the nature and character of the patterns of the stratified hierarchy; and (*b*) modes of recruitment (and the extent to which these reflect social origins and educational experience) and the opportunities or otherwise for mobility and movement between positions (including the degree of upward movement from the ranks). Moreover, in this latter aspect, as a general rule it would appear to be the case that the more bureaucratic and technical the means for the selection of key personnel, the less likely it is that rigid ascriptive norms apply on entry into the occupation; although, *pari passu*, the opportunities for subsequent movement from humble maritime roles into formally accredited positions of decision-making responsibility may be commensurately circumscribed.

In the British situation, however, considerable variations in stratification may be observed in different maritime occupations. Moreover, the greater the extent to which criteria of performance obtain, the less would seem to be the opportunities for the genesis and maintenance of fixed types of social hierarchy. Insofar as the Royal Navy is concerned, therefore, although there are no parallel studies with the work of Otley on army officers [29], Dandeker has produced a particularly interesting account of the patterns of change in an occupation [30]. In the period 1780–1815, a system of patronage substantially influenced selection into high-ranking positions and careers. The sources of naval patronage were both internal and external to the service and encompassed, in descending order of importance: the lords of the admiralty, and other members of the government; the court and landed families well-connected there; members of parliament; serving flag officers; and unemployed but otherwise influential flag officers. In the initial stages, therefore, entry depended upon the entitlement of the commanding officer of a ship commissioned by the admiralty to select 'young gentlemen' into the service as prospective officers, while most of the training consisted of learning 'seamanship skills' rather than any formal body of knowledge. Moreover, because after the crucial rank of captain (via lieutenant and commander) the *de facto* rule of seniority applied, it was essential to enter the service at an early age. This state of affairs was ultimately transformed by the bureaucratization of the service occasioned by an increasingly central mode of control over the opportunities for and practices of promotion. However, the extent to which social origins and self-recruitment within naval families may still have an effect upon the prospects of attaining higher-ranking positions has been insufficiently examined and researched.

Moreover, the patterns which obtain in the Royal Navy would appear to be markedly different from the experience, say, in fishing occupations. Certainly, in Tunstall's account, it would seem that the skipper's position is extremely precarious. Similarly, although the same author found that the bulk of skippers had fathers who had at one time or other been fishermen and a third who had occupied that position itself, the social origins of the bulk of this group would appear to be almost entirely working class [31]. This situation is facilitated by the commercial relationship between shipping company and skipper, since the maintenance of a position (particularly in a declining industry) depends upon a continuing high level of performance measured in terms of the profitability of catches. Hence, while skippers typically were found to have

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moved to more exclusive parts of cities and had material styles of life which sharply separated them from the rest of the crew; they remained, in Tunstall's words a 'super-tax proleteriat' who faced the continual prospect of an early and enforced return to their social roots [32].

The patterns of stratification on board various types of ships have also been substantially affected by technical change. This is manifested not only in the educational level of recruits, but also in the emergence of differing roles and responsibilities amongst the main social groups. To be sure, some occupations implicated here have been traditionally marginal. For example, the marine radio-man has typically regarded himself as roughly on a par with staff officers, an issue of some importance since, as Record has noted, the merchant marine has a hierarchical tradition divided according to occupational and status groupings [33]. At various times, however, trade union action has substantially blurred identities and reference groups and produced a variety of cross-cutting loyalties. Furthermore, as Mann has noted, deck officers have traditionally enjoyed a higher status than engine officers and partly because of this division, although the radio officer is a specialist who works alone (and importantly in some instances is hired from wireless companies), the most usual situation would appear to be to seek contacts and sociability with deck officers [34].

Yet, although this is an under-researched area, the advent of advanced technology has almost certainly had substantial repercussions for decision-making and for informal status-hierarchies on board ship. After all, to the extent that decisions have increasingly come to reflect technical criteria rather than personal judgements based on experience, officers in direct charge of the technological apparatus must have far higher levels of decision-making power than was once the case. Moreover, although this tendency has to some extent been counterbalanced by the higher levels of general education of the 'line officers', in the last quarter of a century in particular, the patterns of authority and status have undoubtedly been subject to substantial pressures for adaptation and change and, in some circumstances, for a radical restructuring of the administration and managerial control system itself. Smith and Roggema have emphasized that increasingly sophisticated technology (together with a competitive trading environment and changing values) have tended to encourage stratified responsibility, a matrix form of organization and decentralized decision-making on board ship [35].

7. Processes

The analysis of social structure has typically been supplemented by an elaboration of the processes involved in interactive relationships. This has been observable in sociological studies of maritime phenomena in two main respects; first in the various models based on exchange analysis (including input-output dimensions); and, second, in the focus on patterns of social interchange in recruitment and occupational socialization.

To begin with, therefore, writers such as Prattis have deployed Barth's theory of transactions involving a recognition of social roles and the exchange of goods and services between the occupants [36]. Moreover, although it could be objected that the cruder versions of exchange analysis have failed to accommodate notions of disparities in power between the role incumbents, that they have been based on an oversimplified conception of dyadic processes and that they have not involved any real acknowledgement of the multidimensional character of social status, a wide range of fairly sophisticated exchange models have evolved in mainstream sociology [37]. In Prattis's analysis a theory of transactions has been married with the notion of status

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incongruency (i.e. the view that different dimensions of status are not always homogeneous or consistent across a range of social relationships) in order to reinforce the case that tensions and discomfort can arise in transactions amongst 'status-incongruent' individuals in maritime life. More specifically, too, in such exigencies, various strategies of 'impression management' may be deployed by those experiencing status incongruency in order to change the evaluation of one or more of the various dimensions and to redefine the situations in which the respective actors are placed [38]. The conflict between shipboard and shore-based status is thus a particularly acute problem in maritime milieux and may be responsible for a series of ambiguities over status and social position in the respective contexts.

An interesting variant of the exchange approach, however, involves identifying a series of 'pull' and 'push' factors affecting decisions to join maritime occupations and, indeed, to remain in service. Mannheim and Rosenstein [39] (following Aubert and Arner [40]) have thus delimited the principal factors as follows:

(*a*) Pull forces to sea: early occupational choice, job-specific maritime occupation, satisfaction with occupational rewards, positive attitudes to ship life, good relations with supervisors.
(*b*) Pull forces to shore: high level of non-maritime education, marriage and parenthood, high economic attractions, good shore-job alternatives.
(*c*) Push forces from sea: mature age, lack of female companionship, dissatisfaction with occupational rewards, negative attitudes to ship life, poor superior-subordinate relations.
(*d*) Push factors from shore: family problems, low economic attainments.

Naturally, in some respects, this reflects a check-list of factors rather than a theory of flows between occupations or a detailed examination of social processes, but it has obvious parallels with exchange modes of analysis and has considerable practical utility.

Indeed, various 'job choice' paradigms have been frequently associated with attempts to delineate processes involved in actual recruitment into maritime occupations and the patterns of socialization therein. Rather than focusing upon structural relationships entailed in recruitment, therefore, it has been frequently observed that training can affect the chances of any individual remaining in a given occupation [41]. Hill, in particular, explicitly studied factors affecting 'wastage' in a seafaring career in order 'to provide information about the characteristics of different types of sea-going careers and the factors affecting length of stay in the industry' and explored 'the alternative courses of remedial action which the industry might undertake' [42]. By the same token, in studies of fishermen, the factors affecting decisions to join the occupation and to remain have been examined at various levels including skippers and deck-hands [43].

Nevertheless, while analyses of social processes have involved useful check-weights against the crudest versions of structural determinism, in future accounts, a greater conceptual rigour and the integration of specific research into the multidimensional forces and factors involved in the complex modes of occupational socialization would be a worthwhile departure. At present in a maritime context, however, many of the perspectives based on notions of process have been at root rather descriptive in character and have lacked a secure foundation in those penetrating explanatory propositions upon which genuine theories of social interaction amongst the personnel in question should really be constructed.

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8. Change

Dynamic patterns in social relationships are generated not only by processes of interaction but also, more especially, by pressures for change. In any given social system these are usually classified into *endogenous* forces (including technology and social values) and *exogenous* elements (arising from wider physical, political, legal, economic and social movements). Hence, in a maritime context these twin characteristics may be classified as follows:

(*a*) Endogenous changes:
- (i) Alterations in technology: the impact of advanced transportation and extraction techniques on the socio-technical systems.
- (ii) Changes in social values: adaptations in ethics, ideologies and in more specific objectives and purposes *within* the various maritime occupations.

(*b*) Exogenous changes:
- (i) Physical: location of resources (fish, oil and so on).
- (ii) Political: strategic implications of changes in policies on, say, fishing rights or international conflict and trade. Control by external authorities over decision-making in maritime institutions.
- (iii) Legal: implications of enactments on maritime issues.
- (iv) Economic: general economic conditions and their impact on maritime industries. The more specific situation of maritime concerns themselves.
- (v) Social: changes in ethics, ideologies and values in the wider society and culture.

Insofar as the endogenous changes are concerned, therefore, we have already mapped out the probable effects of technology for patterns of social stratification on board ship that, in turn, may be seen to stem from alterations in skill level and expertise of the personnel in question. Nevertheless, a number of further implications are worth tracing at this point and notably the change from community to association, patterns of work and leisure, and the tendency to encourage instrumental types of orientation to work. Hence, although as we have seen that, in maritime milieux, it has been usual to view community as a basis for association; equally the changes in skill level and the more heterogenous tasks required in modern vessels tend to undermine some of the earlier bonds based on cohesive occupational loyalties. Moreover, containers and other advances in transportation systems, organizational changes (such as growing size of units) and rapid turn-round times of ships in ports have all clearly had important ramifications for work and leisure in both ship-board and shore-based contexts. Indeed, as Moreby has observed:

> Although the principles of naval architecture, thermodynamics and electricity have remained unaltered for 100 years, the ways in which these principles are being applied are changing rapidly. Time gaps between invention, first commercial use, general use and obsolescence are decreasing; people in the shipping industry are having to cope with significant changes in marine technology [44].

Again, the erosion of occupational and craft-based loyalties may well have encouraged increasingly instrumental attitudes to work in which extrinsic benefits associated with wages and conditions gradually supersede earlier types of identity of the personnel themselves [45]. But the members of the various maritime occupations are also, of course, social actors who, quite independently of technical change, can generate their own attitudes in ways which encapsulate the emergence of fresh ethical or ideological premises within the occupations in question.

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Furthermore, changes in maritime environments are often occasioned by exogenous forces but still have far-reaching and highly significant consequences for internal processes of decision-making and patterns of interrelationships in the occupations concerned. Alterations in the location of resources (and particularly fluctuations in fish stocks) have thus historically had fundamental repercussions for fishing communities. Again, wider political decisions on rights and international relationships have had profound implications for maritime personnel. Similarly, legislative intervention on safety standards, minimum conditions for employment and so on all clearly affect social patterns on board ship and result in substantial modifications in existing procedures. General economic conditions, too, can obviously affect the well-being of a given maritime industry and, on an indirect basis, the material resources available, say, for the naval fleet. And, finally, wider changes in social values, ideologies, ethics etc. in a given socio-cultural system can all bear upon the more specific patterns and affect the degree of formality of procedures, the treatment of various groups on board ship, the willingness to join trade unions and so on. Thus, as Moreby has stressed:

> Seafarers are recruited from shore communities to which they return periodically on leave. To some degree, seafarers are affected by changing values in shore societies; they import these values into the shipping industry and expect the values of the shipboard society to harmonise with shore-based values. Changing social values cannot be ignored by those engaged in recruiting, training and employing modern seafarers [46].

Hence, although the components and the effects of social change are improperly understood and are certainly under-researched in maritime contexts, they clearly form an essential part of a comprehensive analysis of this emerging area of study.

9. Conflict and deviance

The analysis of social phenomena in maritime locales has also encompassed attempts to map out and to explain conflictual aspects of relationships and the engagement or otherwise of the personnel in deviant modes of action and behaviour. Nevertheless, the familiar lament that this approach has evolved in an *ad hoc* or grounded fashion, rather than being anchored in the precepts and categories of the parent discipline, has again considerable force in this particular context. Indeed, broadly speaking, while the types of conflict which have been identified in maritime life are clearly many and various, the principal groupings which have been isolated may be adumbrated as follows:

(*a*) Extra-organizational patterns: with companies, governments and competing organizations; trade-union activities directed against such bodies.

(*b*) Intra-organizational patterns: between the different stratified groups in the main organizations, trade-union activities representing the section in question.

(*c*) Individual responses; unorganized types of conflict such as high rates of labour turnover and 'wastage rates', absenteeism and accidents, sabotage of machinery and equipment.

(*d*) Socio-psychological aspects; patterns of disharmony with the family and individual members of the community. Problems of adaptation and socialization including disillusionment and disengagement. Mental breakdown, alcoholism, sexual deviancy, petty crime and so on.

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To begin with, therefore, conflict in maritime life may be directed against external groupings and involve all members of an organization in some collective action against these wider bodies. For example, the whole crew of a fishing vessel may be hostile to a shipping company, while in the navy this animosity may be directed to a government embarking upon financial cuts or bureaucratization of the service. In this latter aspect, Dandeker noted how the 'impersonalization' of rules, the specialization and permanence of positions and the decentralization of specified authority were all developed from governmental initiatives and that this involved a certain degree of conflict with existing personnel who were wedded to rather different practices [47]. Equally, however, there may well be internal sources of conflict and disharmony centred around questions of power, authority and specified competence, which involve not just relationships between officers and crew, but also those between members of different functional segments (for example, deck and engine departments). Part of this may stem from technical change but, as we have also noted, many types of conflict arise when new expectations about appropriate relationships come to the fore. Moreover, in response to such exigencies, the growth of ship-board liaison schemes, of trade-union activity on the actual vessels, and a variety of modes of employee participation may all be seen as responses to such patterns of internal conflict [48].

There is also a substantial body of literature in which individual reactions to problems in a work envirionment have been surveyed and assessed. Certainly the various contributions to the analysis of labour turnover, absenteeism, accidents and 'wastage' have been familiar landmarks [49]. Moreover, in further approaches, various types of rule-breaking behaviour including the sabotage of equipment and machinery have been examined [50]. Again, it has been fully appreciated that processes of occupational socialization and training are by no means wholly satisfactory and frequently involve substantial problems of adaptation for the new recruit and, in certain cases, disillusionment and disengagement [51]. Moreover, these individual socio-psychological responses may frequently take the form of deviant modes of behaviour including mental breakdown, alcoholism, abnormal sexual behaviour and petty crime [52].

Nevertheless, while the literature on conflictual and deviant modes of conduct in maritime life is considerable, as we have observed, this has seldom been anchored in those sociological categories which have been familiar in general debates and modes of explanation themselves. Certainly both the 'structural' and 'subjective' sources of conflict have been improperly understood, while deviant types of reaction have seldom been couched in broader concepts including 'labelling theory', alienation, anomie and so on [53]. Hence, while it is indisputable that the study of maritime phenomena would be incomplete without some reference to these conflictual aspects of social relationships, in forthcoming contributions, a more rigorous and explanatory mode of analysis grounded in the precepts of the parent discipline would be invaluable.

10. Conclusions: a strategy for research and conceptual advance

In the foregoing analysis we have attempted to evaluate the main areas of focus within maritime sociology and to link the various substantive discussions more closely than hitherto with general theoretical and conceptual approaches. It remains to recapitulate those themes to which we attach a special significance, while examining the main components of a future strategy for research and for abstract inquiry.

Within the compass of the sociological literature on maritime studies, therefore, seven major issues have proved to be particularly influential in channelling and

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directing the attention of the analyst: (*a*) the concept of a social system applied to the social structure and crew of ships (including the notion of total institutions); (*b*) community and association; (*c*) social action and culture; (*d*) structure, differentiation and stratification; (*e*) process (encompassing patterns of recruitment and socialization); (*f*) change (including the effects of the introduction of new technologies); and (*g*) conflict and deviance. Again, in each and every case, a variety of sub-issues and themes have also been isolated for special consideration. Furthermore, in the course of our own review, particular attention was drawn to the considerable importance of developing research designs in which central concepts are explicitly incorporated and tested within rigorous explanatory models. Such a view, too, was advocated as a check-weight against the bulk of the analyses deployed so far because, in the great majority of cases, these have emerged as grounded theories bearing very little relationship to the accumulated body of knowledge within sociology itself.

More specifically, however, on the basis of our survey of themes and approaches, it would seem reasonable to infer that research in maritime sociology would benefit from the following developments in substantive focus, theoretical perspective and research technique:

(i) An extension of the groups covered in maritime social analysis to include those engaged in gas and oil exploration and extraction together with various occupations (such as lifeboat crews and lighthouse operations) which have not as yet featured prominently in the literature. Moreover, these studies should include some account of the attitudes and behaviour of female as well as male employees [54].

(ii) In the case of officers, a detailed examination of social origins and educational background as well as career paths and job mobility should be undertaken. Studies of job satisfaction, too, should cover such groups as well as those of maritime operatives. This would supplement the work on officer behaviour [55].

(iii) Insofar as crew members themselves are concerned, future analysts might well find it worthwhile to examine perceptions based on Weberian precepts and also to attempt to integrate structural and action models.

(iv) The deployment of more detailed studies of interaction based on socio-psychological propositions and research techniques would also be invaluable.

(v) More generally, too, explanatory modes of analysis should be formulated and should transcend the mere description of salient characteristics of maritime occupations.

(vi) Similarly, the evolving explanatory models should be comprehensive in compass and involve not only the examination of phenomena based on the notions of structure, culture and action, but also the role of economic conditions, legal and governmental intervention, and technical change.

(vii) Multivariate research designs should also be developed and should entail the use of correlation and multiple-regression techniques to isolate and to attach weights to strategic variables.

In sum, therefore, if maritime sociology is to become a sophisticated area for study and research, it has to advance beyond the confines of grounded theory and to reflect the advances in theory, methodology and research design that have eventuated in the social sciences in the last decade. However, given such a change in focus and emphasis,

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there is no reason to suppose that, in this area of study, more exacting standards of analytical and conceptual rigour will not ultimately be forthcoming and in the process ensure that this will become an exciting context for the testing of different explanatory models and propositions. After all, there is little doubt that, whatever the vicissitudes of contemporary economic and political movements, this aspect of social life has in no way lost its substantial appeal as a source for both scholarly and popular disputation.

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